

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

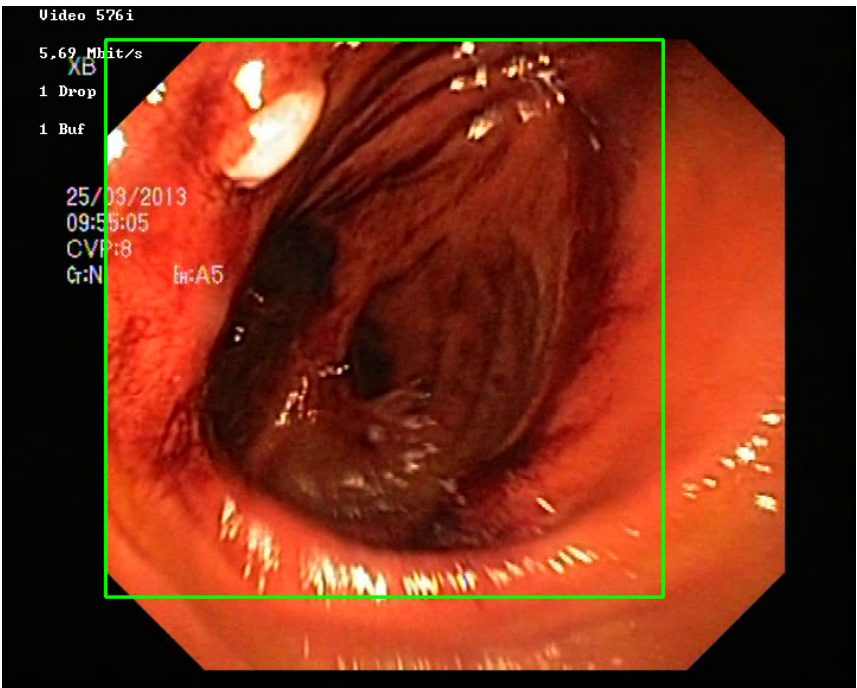
Video ID:	8f9e4085-98c4-4f69-99e6-3945dda43c05
Total Frames:	2638
Frame Rate:	25.0 fps (assumed)
Duration:	105.5 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	171	✓
RT-DETR Detections	253	✓
MedSAM2 Detections	374	✓
Consensus (3-Model Agreement)	3	✓
Risk Classification	Count	Percentage
HIGH RISK	1	33.3%
MEDIUM RISK	0	0.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	3	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 17 frames)



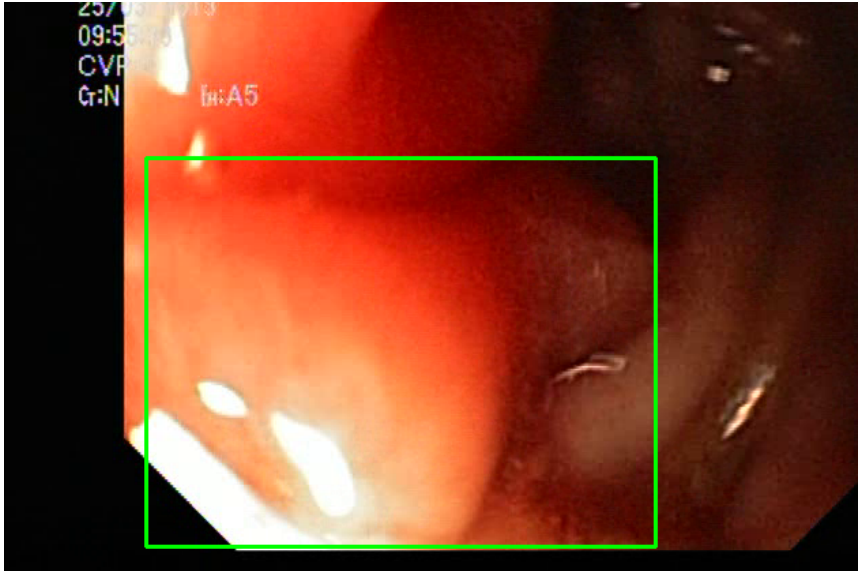
Feature	Value	Interpretation
Frame Index	858	Frame 858 of 2638
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Detector Consensus (MedSAM2 Unavailable) (MedSAM2 Unavailable for this video)
POLYP TYPE	BLEEDING_POLYP	Type confidence: 92.0%
Redness Score	0.467	Color-based vascularization
Relative Radius	43.7%	Polyp size as % of frame diagonal
Texture Score	0.526	Surface morphology
Vessel Visibility	0.874	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	MALIGNANT_POLYP	High suspicion for malignancy — biopsy and urgent resection recommended

Detection Confidence	71.6%	YOLO / RT-DETR / track confidence
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Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #2 — MALIGNANT_POLYP [HIGH_RISK] (Tracked across 23 frames)



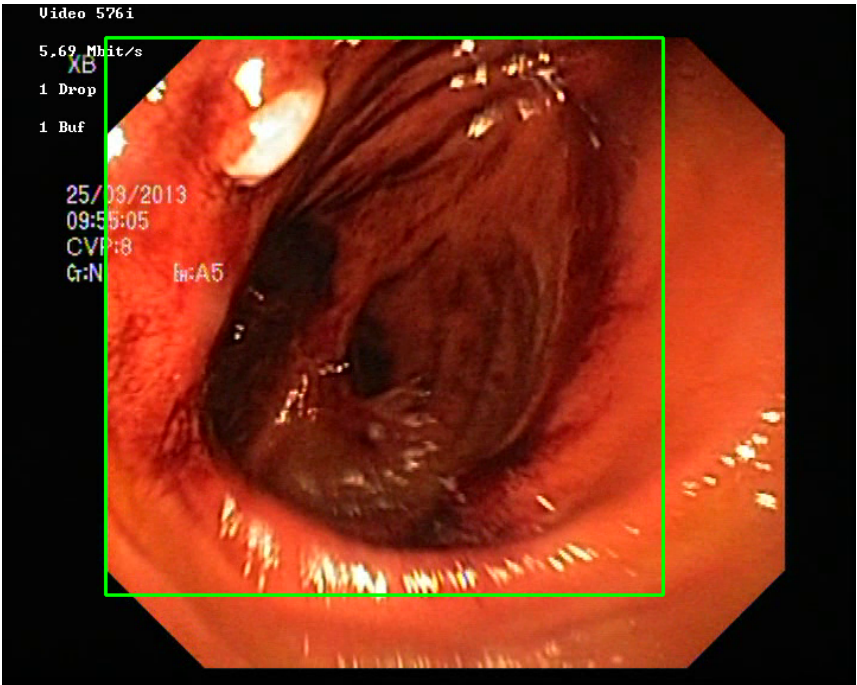
Feature	Value	Interpretation
Frame Index	1552	Frame 1552 of 2638
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	MALIGNANT_POLYP	Type confidence: 92.0%
Redness Score	0.357	Color-based vascularization
Relative Radius	50.2%	Polyp size as % of frame diagonal
Texture Score	0.084	Surface morphology
Vessel Visibility	0.567	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket

Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	65.9%	YOLO / RT-DETR / track confidence

Type Assessment: Morphology consistent with high-grade neoplasia — biopsy and urgent resection strongly recommended.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #3 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 17 frames)



Feature	Value	Interpretation
Frame Index	858	Frame 858 of 2638
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	BLEEDING_POLYP	Type confidence: 92.0%
Redness Score	0.467	Color-based vascularization
Relative Radius	43.7%	Polyp size as % of frame diagonal
Texture Score	0.526	Surface morphology

Vessel Visibility	0.874	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	MALIGNANT_P OLYP	High suspicion for malignancy — biopsy and urgent resection recommended
Detection Confidence	59.0%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

CLINICAL IMPRESSION

Summary:

A total of 3 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 1 (33.3%)
- MEDIUM RISK polyps: 0 (0.0%)
- LOW RISK polyps: 0 (0.0%)

Recommendation:

The presence of HIGH RISK polyps warrants close clinical attention. Consider endoscopic resection or follow-up surveillance depending on clinical context.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.